

CONTACT

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EDUCATION

PhD, Computer Science (part-time)
Indian Institute of Technology Delhi
Aug 2025 – present

M.Tech, Computer Science & Engineering
Punjab Engineering College, Chandigarh
2019 – 2021 · CGPA 9.60 / 10

B.Tech, Computer Science & Engineering
Punjabi University, Patiala
2015 – 2019 · CGPA 8.21 / 10

SKILLS

DATA & BACKEND

Python · SQL · Apache Airflow · Elasticsearch · MySQL · FastAPI · Drupal · PHP

PLATFORMS & APIS

Kubernetes · Apache APISIX · REST APIs · DCAT · Git

AI / ML

Machine learning · CNNs & GANs · LLMs & RAG · Clustering

GEOSPATIAL

Google Earth Engine · Sentinel-1 / 2 · Remote sensing

LEADERSHIP

Team leadership · Capacity building · Stakeholder engagement

Jaskaran Singh

Scientist-C / Deputy Director · National Informatics Centre, MeitY
Part-time PhD Scholar · Computer Science, IIT Delhi

PROFILE

Technology leader at the National Informatics Centre working on data.gov.in, India's national Open Government Data platform, with government departments, states, academia and industry. Part-time PhD scholar at IIT Delhi researching satellite-based forest biodiversity monitoring; previously a data scientist at McKinsey & Company.

EXPERIENCE

National Informatics Centre · MeitY, Government of India — New Delhi

Scientist-C / Deputy Director — Aug 2026 – Present
Scientist-B / Assistant Director — Mar 2022 – Aug 2026

Bharat Data Platform — data.gov.in, the Open Government Data Platform of India

- Head a multidisciplinary team that designs, builds and scales the platform; mentor the team and state partners on modern technologies, coding practice and data-publishing standards.
- Designing the next-generation platform: microservice decomposition, DCAT-3 metadata, an Apache APISIX API gateway, and disaster-recovery and backup planning.
- Build Apache Airflow pipelines on Kubernetes, including scheme feeds such as Jal Jeevan Mission and a monthly, checkpoint-resumable metadata-enrichment job, designed for lakhs of resources, that flags changed records for search re-indexing.
- Catalog and API clean-up programme (a unified Elasticsearch index and endpoints for the PMU): built a snapshot-diff tool that tracks each catalog's state (linked, orphaned, absent) across database snapshots.
- Work on platform services: the visualisation API (PHP, MySQL), a FastAPI PII-detection service, the Feedback Management System, and the Elasticsearch cluster.

McKinsey & Company · Operations Digital Assets team — Gurugram

Data Scientist — Jul 2021 – Mar 2022

- Developed pipelines for internal tools aimed at reducing the time client engagements take, and was responsible for developing and deploying tools for firm-wide access.
- Worked on data extraction, cleaning and processing, and on developing and deploying models, across client engagements.

Intern — Jan 2021 – Jul 2021

- Built a system that takes documents of any type and extracts structured information for analytics; implemented a layout-extraction algorithm that improved the firm's knowledge-extraction tools, and developed a working prototype for demonstration.

RESEARCH

PhD, IIT Delhi · forest biodiversity monitoring from satellite data — Aug 2025 – Present

Thesis (working title): A Computational Framework for Forest Biodiversity Monitoring and Modeling · Supervisor: Prof. Aaditeshwar Seth

- **Forest stand delineation.** Built an unsupervised multi-sensor pipeline in Google Earth Engine (Sentinel-2 phenology, Sentinel-1 SAR, canopy height, terrain) using SNIC superpixels and k-means; tested at Sanjay Van (Delhi) and Mudumalai Tiger Reserve.
- **Foundation-model embeddings.** Extended the pipeline to cluster on AlphaEarth Foundations embeddings (Google DeepMind) as an alternative to hand-crafted features; a comparison of the two is planned.
- **Forest condition for rural planning.** Collaborating with the Foundation for Ecological Security on a forest-condition decision layer for rural-employment planning under VB-G RAM G; pilot scope Odisha and Rajasthan.

CERTIFICATIONS

Microsoft Certified: Azure Data Fundamentals
Microsoft

Machine Learning with Python – Level 1
IBM

Cloud Core
IBM

Capstone: Retrieving, Processing, and Visualizing Data with Python (with Honors)
University of Michigan · Coursera

Using Databases with Python
University of Michigan · Coursera

LANGUAGES

Punjabi	Native
Hindi	Native
English	Professional working

INTERESTS

AI / ML · NLP · Geospatial analytics · Open data · Microservices · Data security · IoT

Jaskaran Singh

Scientist-C / Deputy Director, NIC · PhD Scholar, IIT Delhi

SELECTED PROJECTS

Offline question-answering assistant for data.gov.in — Personal project

- Retrieval-augmented chatbot that runs fully on a laptop (Ollama, sentence-transformers, ChromaDB, FastAPI with streaming). A moderation gate and intent classifier route queries to document search, live-data lookup or request filing; answers are grounded in retrieved text and out-of-scope questions are declined.

Breast-cancer detection from mammograms — Jan 2020 – Jun 2020

- Convolutional neural network classifier with tuned hyper-parameters; trained a GAN to generate synthetic mammograms that augment the original dataset.

INTERNSHIPS

Sabudh Foundation
SAS Nagar, Punjab
Data Science Intern — Jul 2020 – Dec 2020

TheSmartBridge · SmartInternz
Patiala · remote
Internship project — Apr 2020 – May 2020

- Developed an intelligent customer help desk with smart document understanding.

Infosys
Mysuru
Systems Engineer Intern — Jan 2019 – May 2019

- Built an email server from scratch, secured with salted hashing and AES-256 encryption, with QR-code access to details for administrators; shared its API with a peer team's movie-ticketing project, the first team to combine two projects, which was recognised by mentors.

LEADERSHIP

CDGC — Training & Placement Cell · PEC Chandigarh
Joint Head (M.Tech) — Aug 2020 – Jul 2021
Training Class Representative (M.Tech) — Jan 2020 – Aug 2020

- Coordinated between students, faculty and the placement cell on recruitment and industry engagement; helped organise placement drives, internships and corporate outreach, and represented the M.Tech cohort in policy discussions.